

One hot Encoding

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Document #1

He is a good boy. She is also good.

Document #2

Radhika is a good person.

Vocabulary

a, also, boy, good, He, is, person, She, Radhika

	a	also	boy	good	He	Is	person	She	Radhika
Index	0	1	2	3	4	5	6	7	8

Assign index for each word in vocabulary

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Document #1

He is a good boy. She is also good.

Document #2

Radhika is a good person.

Vocabulary

a, also, boy, good, He, is, person, She, Radhika

	a	also	boy	good	He	is	person	She	Radhika
Index	0	1	2	3	4	5	6	7	8
Document #1	1	1	1	2	1	2	0	1	0
Document #2	1	0	0	1	0	1	1	0	1

Document as Vector

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(CountVectorizer)

Word	Index	0	1	2	3	4	5	6	7	8
a	0	1	0	0	0	0	0	0	0	0
also	1									
boy	2									
good	3									
He	4									
is	5									
person	6									
She	7									
Radhika	8									

In one hot encoding ...each word is a as vector

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Word	Index	0	1	2	3	4	5	6	7	8
a	0	1	0	0	0	0	0	0	0	0
also	1	0	1	0	0	0	0	0	0	0
boy	2									
good	3									
He	4									
is	5									
person	6									
She	7									
Radhika	8									

Words as a vector

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Word	Index	0	1	2	3	4	5	6	7	8
a	0	1	0	0	0	0	0	0	0	0
also	1	0	1	0	0	0	0	0	0	0
boy	2	0	0	1	0	0	0	0	0	0
good	3	0	0	0	1	0	0	0	0	0
He	4	0	0	0	0	1	0	0	0	0
is	5	0	0	0	0	0	1	0	0	0
person	6	0	0	0	0	0	0	1	0	0
She	7	0	0	0	0	0	0	0	1	0
Radhika	8	0	0	0	0	0	0	0	0	1

One hot encoding

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How do we use one hot encoding a document.

- Replace each word in the document with its one hot encoding.
- Keep word sequence intact i.e 1st word comes first, 2nd word second and so on.

Document #1

He is a good boy. She is also good.

Document#1	Word Index	0	1	2	3	4	5	6	7	8
He	4	0	0	0	0	1	0	0	0	0
is	5	0	0	0	0	0	1	0	0	0
a	0	1	0	0	0	0	0	0	0	0
good	3	0	0	0	1	0	0	0	0	0
boy	2	0	0	1	0	0	0	0	0	0
She	7	0	0	0	0	0	0	0	1	0
is	5	0	0	0	0	0	1	0	0	0
also	1	0	1	0	0	0	0	0	0	0
good	3	0	0	0	1	0	0	0	0	0

Document as matrix

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Document #2

Radhika is a good person.

Document#1	Word Index	0	1	2	3	4	5	6	7	8
Radhika	8	0	0	0	0	0	0	0	0	1
is	5	0	0	0	0	0	1	0	0	0
a	0	1	0	0	0	0	0	0	0	0
good	3	0	0	0	1	0	0	0	0	0
person	6	0	0	0	0	0	0	1	0	0

Document → 5 x 9 matrix



What are the issues with One hot encoding.

- For large vocabulary size (common in NLP problems), document matrix becomes huge.
- Word vector do not provide any information on how words are related.



Discovering relationships between words

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King + Man = Queen + ?

Can we use **one hot** encoding to solve this equation?

King + Man = Queen + Woman

How do we get numbers which show
relationship between words?

King + Man = Queen + Woman

↙	↘	↙	↘
440	480	530	390
310	410	620	100
220	310	298	222
590	290	352	528
100	517	291	326

...something like this

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Which is similar to 'cat'?



Plane



Bed



Dog



Boy

How can we get word vectors to answer such questions?

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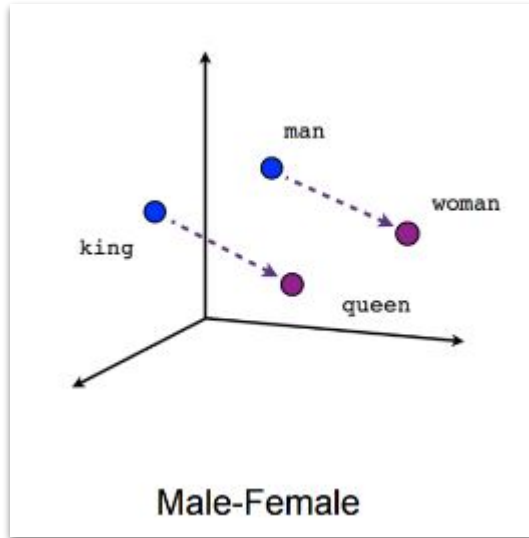
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Discovering Semantic relationship using

Word2Vec

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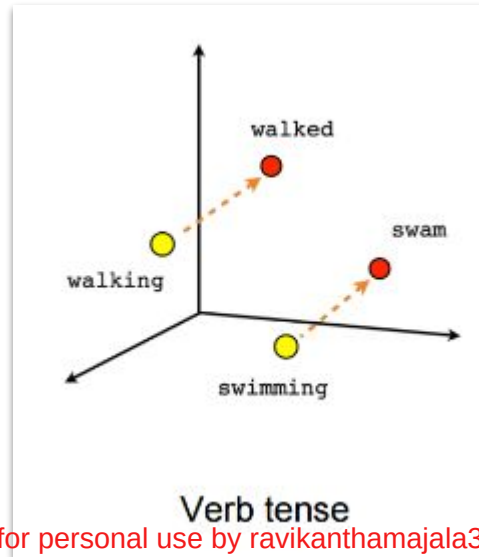
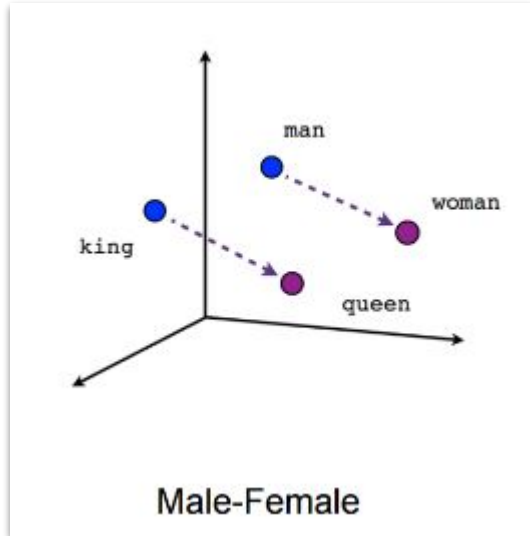


Word2vec provides a vector for each word which can help discover relationship between words

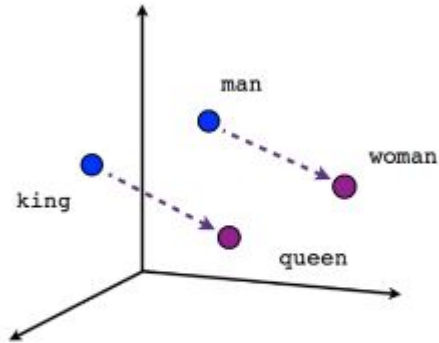
Distance between word vectors of
King & queen

=

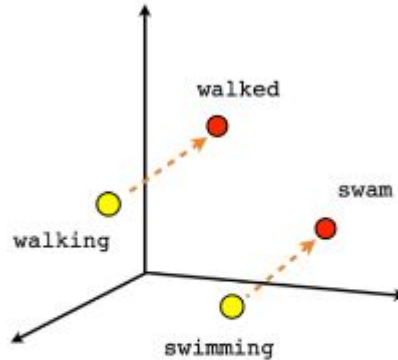
Distance between word vectors of
man and woman



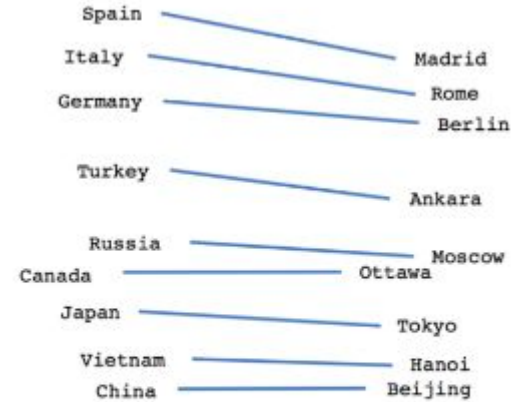
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Male-Female



Verb tense



Country-Capital



How to create Word2Vec vectors?

We use machine learning (ML) to create these vectors. There are two ways to build word2vec embeddings (vectors).

1. Supervised Learning
2. Unsupervised (or Self Supervised) Learning



How does Word2Vec create embeddings using unsupervised learning?

It **learns** word embeddings by understanding word's **neighbours**.

The	Sun	rises	in	the	east
-----	-----	-------	----	-----	------

Given a word, what are the nearby words

The	Sun	rises	in	the	east
-----	-----	-------	----	-----	------

(Sun, The)

Context, Target pair

**For 1st word 'The', who are it's
neighbours**

The	Sun	rises	in	the	east
-----	-----	-------	----	-----	------

The	Sun	rises	in	the	east
-----	-----	-------	----	-----	------

Context, Target pair

(Sun, The)

(The, Sun) (rises, Sun)

**...neighbours of second word
'Sun'**

The	Sun	rises	in	the	east
-----	-----	-------	----	-----	------

The	Sun	rises	in	the	east
-----	-----	-------	----	-----	------

The	Sun	rises	in	the	east
-----	-----	-------	----	-----	------

Context, Target pair

(The, sun)

(Sun, The) (Sun, rises)

(Sun, rises) (in, rises)



What is considered near?

We can decide on the same using a **window size**.

The	Sun	rises	in	the	east
-----	-----	-------	----	-----	------

Window size = 1

(rise, in) (the, in)

**One word to the left and
one to the right are
considered neighbours**

Window Size

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The	Sun	rises	in	the	east
-----	-----	-------	----	-----	------

Window size = 1

(in, rises) (in, the)

Window size = 2

(rises, in) (Sun, in) (the, in) (east, in)

**Two words to the left and
two to the right are
considered neighbours**

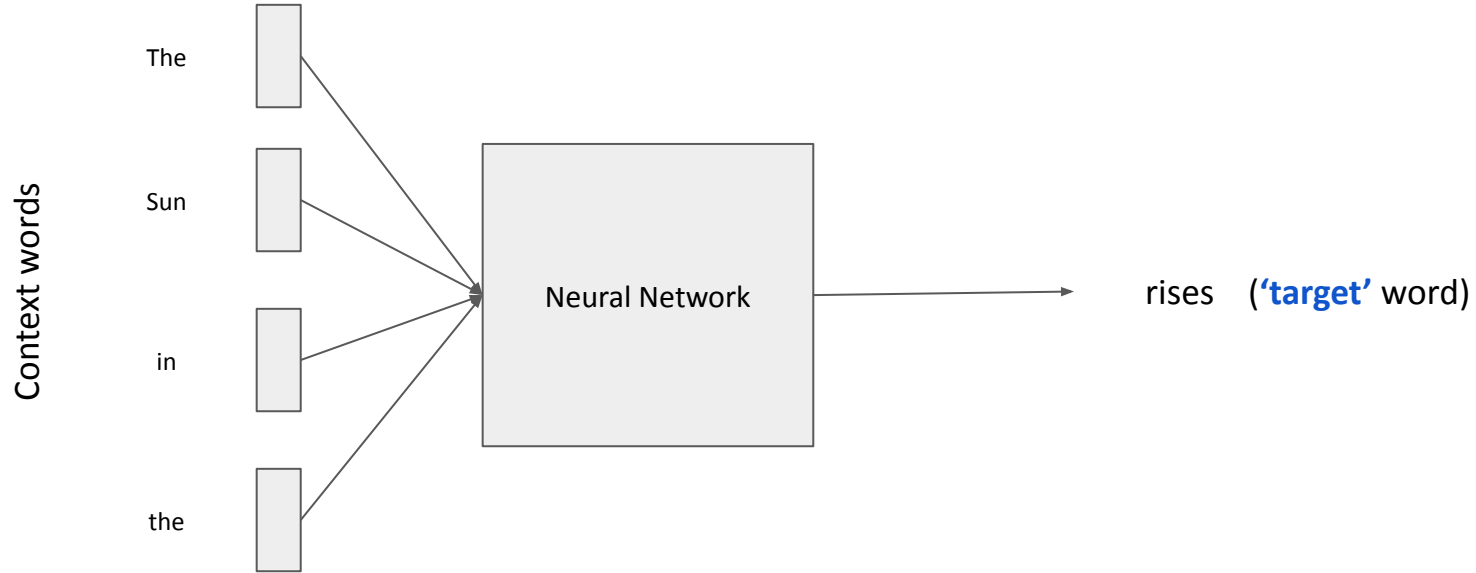
Window Size

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Building Word2Vec Embeddings

2 ways to get it

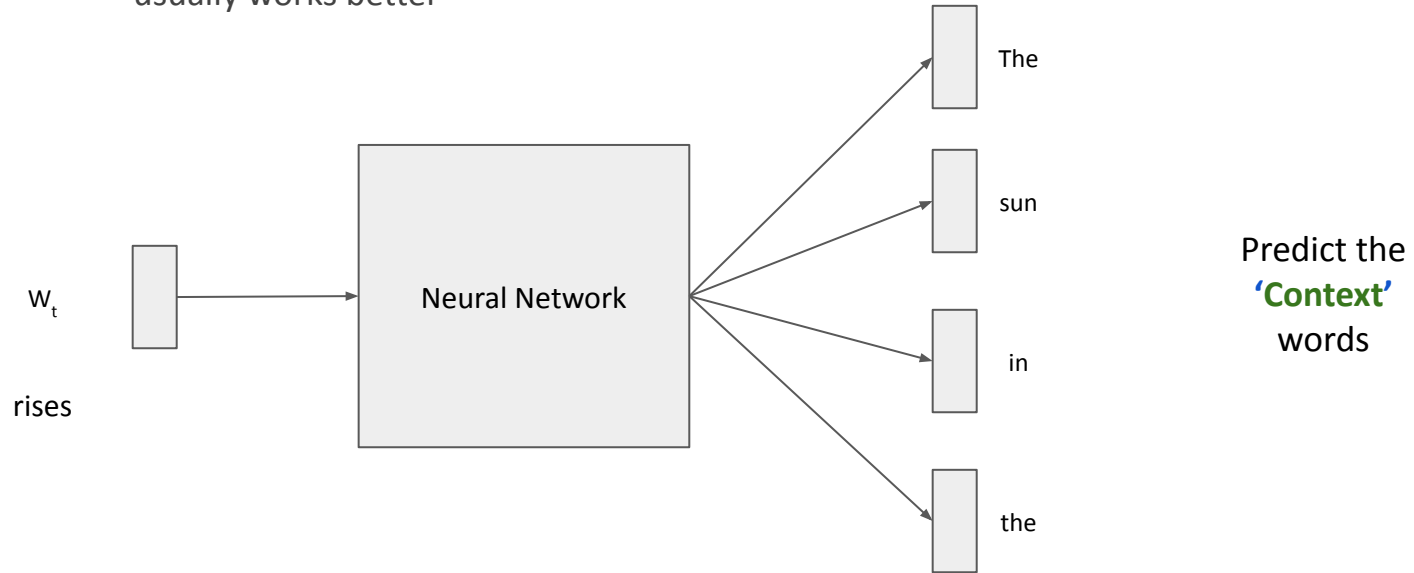
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CBOW model

Skip-Gram model

usually works better



Word2vec model uses a simple Neural network

- 1 Input Layer
- 1 Hidden Layer
- 1 Output Layer

Input word as **one hot vector**

0
1
0
0
0
...
...
0
0

W_t

Input layer

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Input word as **one hot vector**

0
1
0
0
0
0
...
...
0
0

Size of the input
vector?

W_t

Input layer

Input word as **one hot vector**

0
1
0
0
0
0
...
...
0
0

*Same as
vocabulary size*

W_t

Input layer

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Input word as **one hot vector**

[1,10000]

0
1
0
0
0
0
...
...
0
0

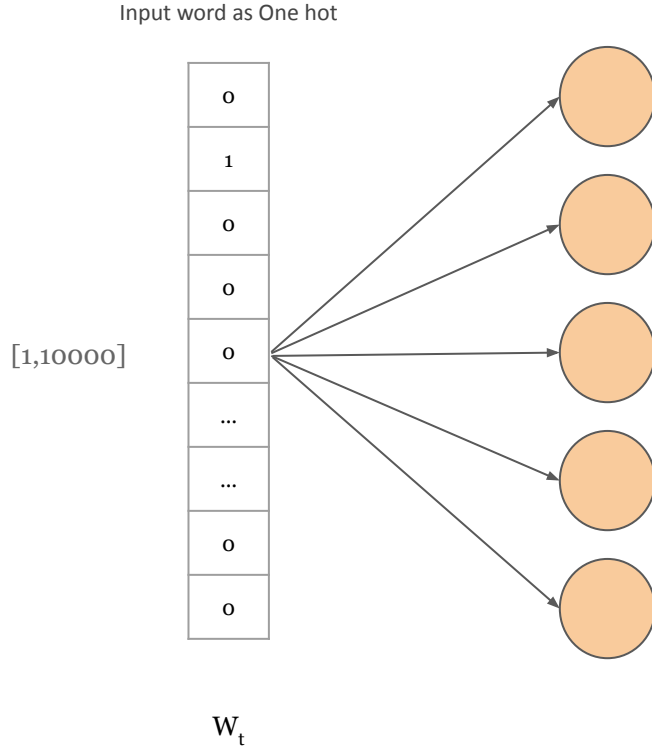
Assume we have
10000 words
vocabulary

W_t

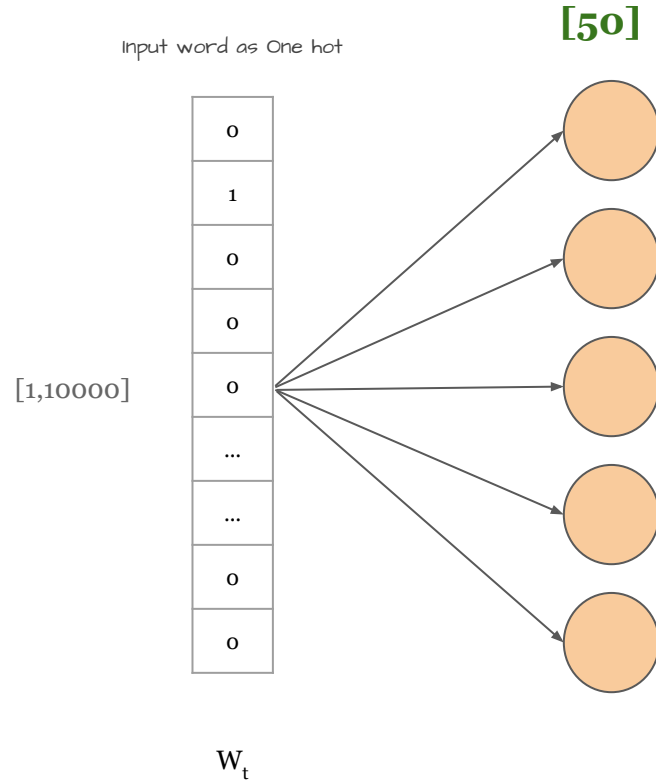
Input layer

Hidden Layer

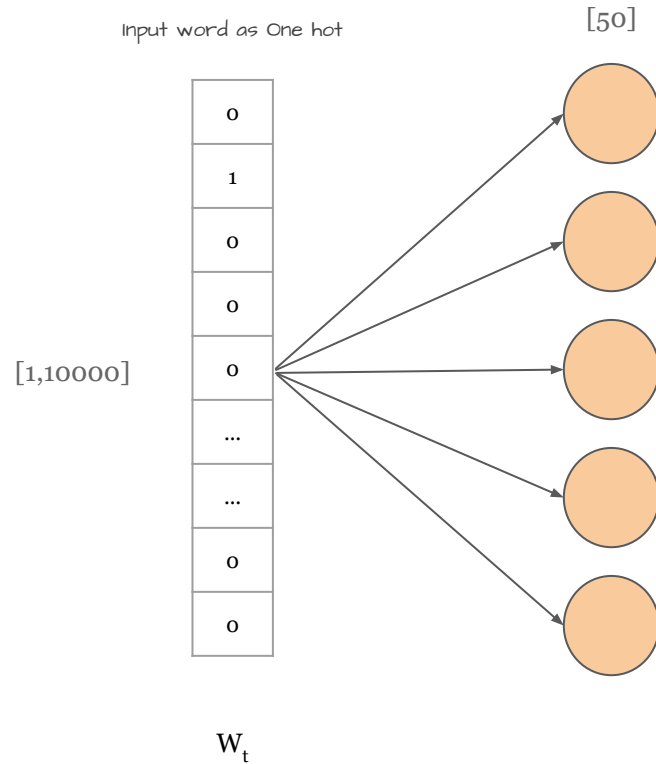
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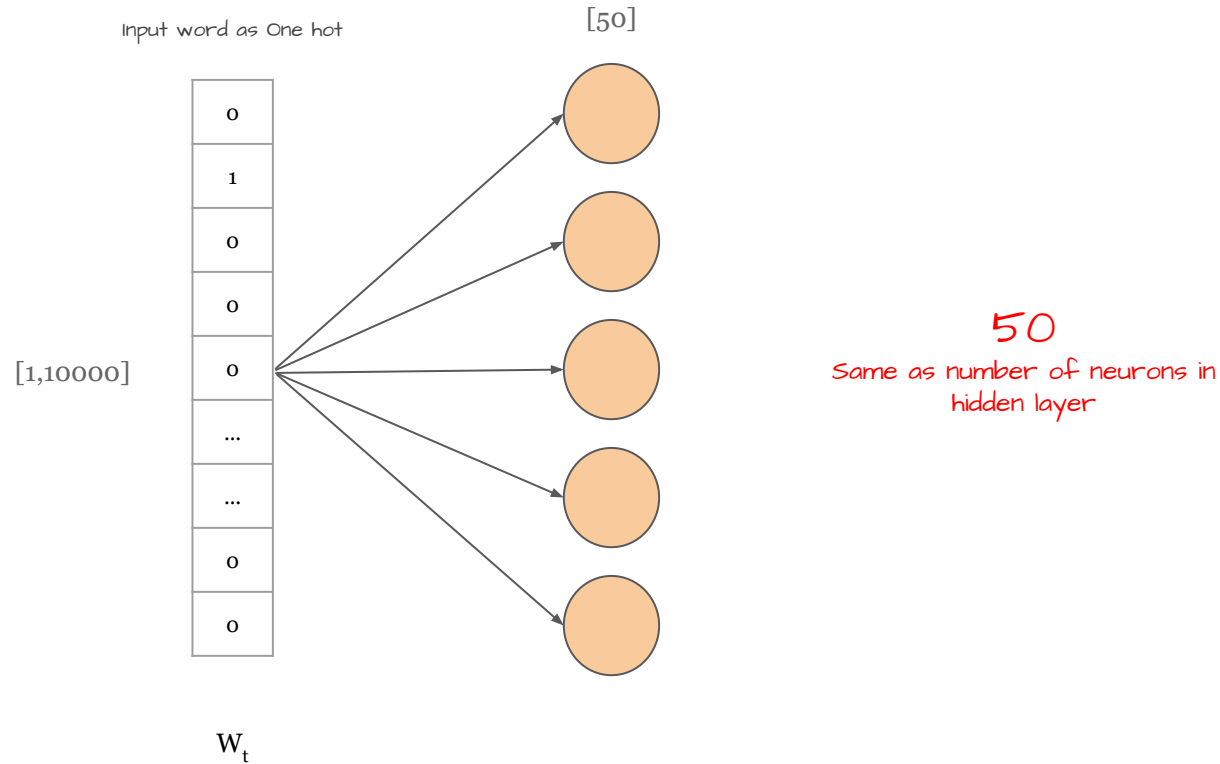
How many
neurons in hidden
layer?



Let's have 50
(or whatever you like)

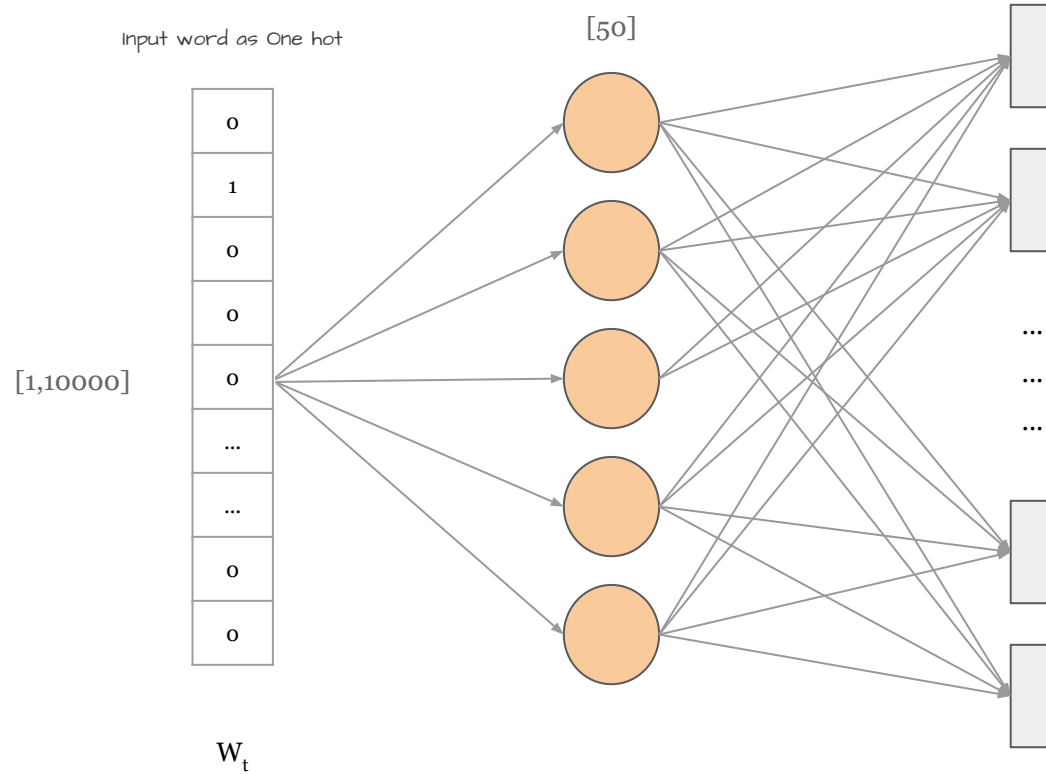


How many hidden
layer outputs
for each Word?

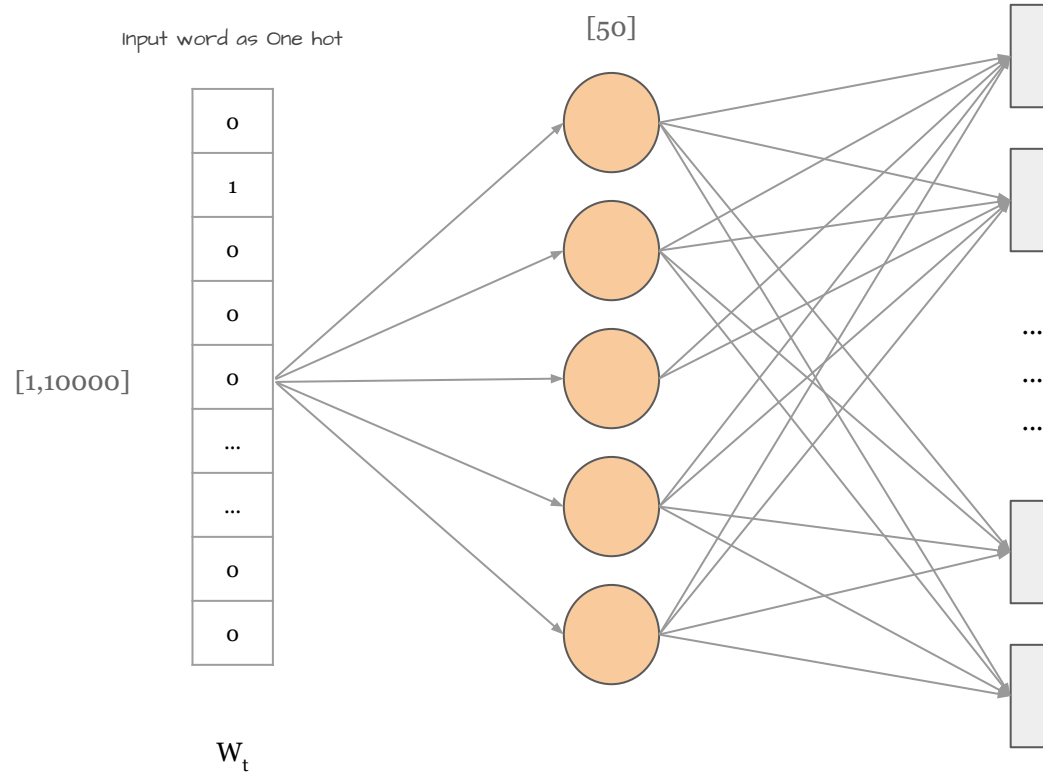


The Output Layer

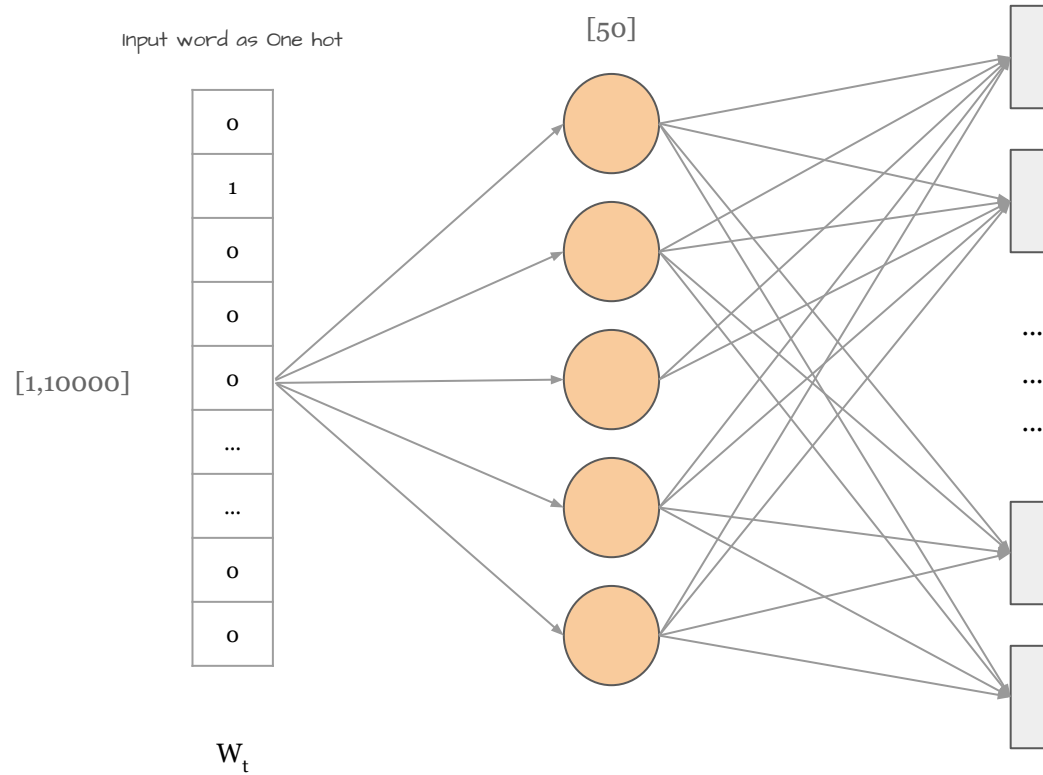
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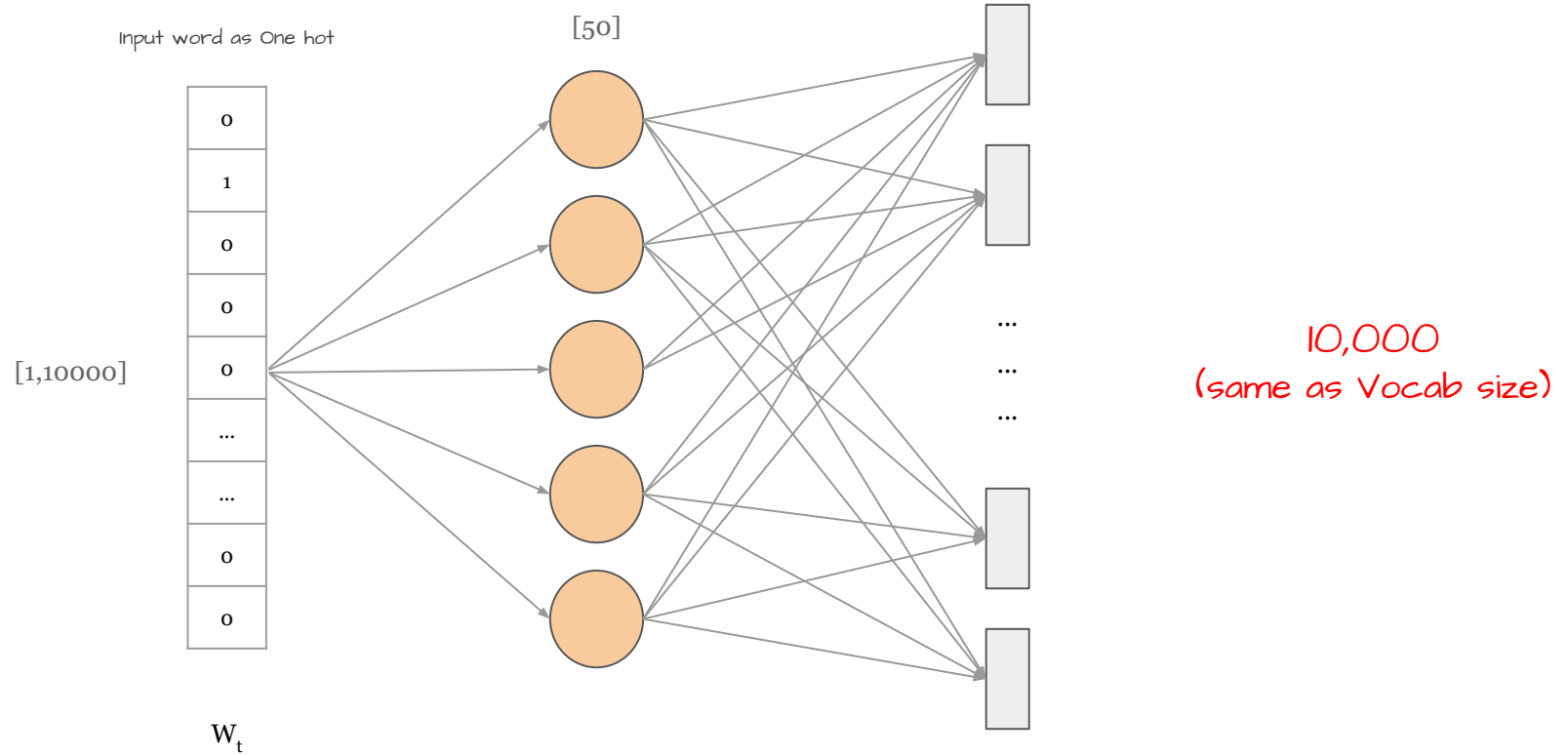
How many
Outputs?



10,000
Same as Vocabulary size

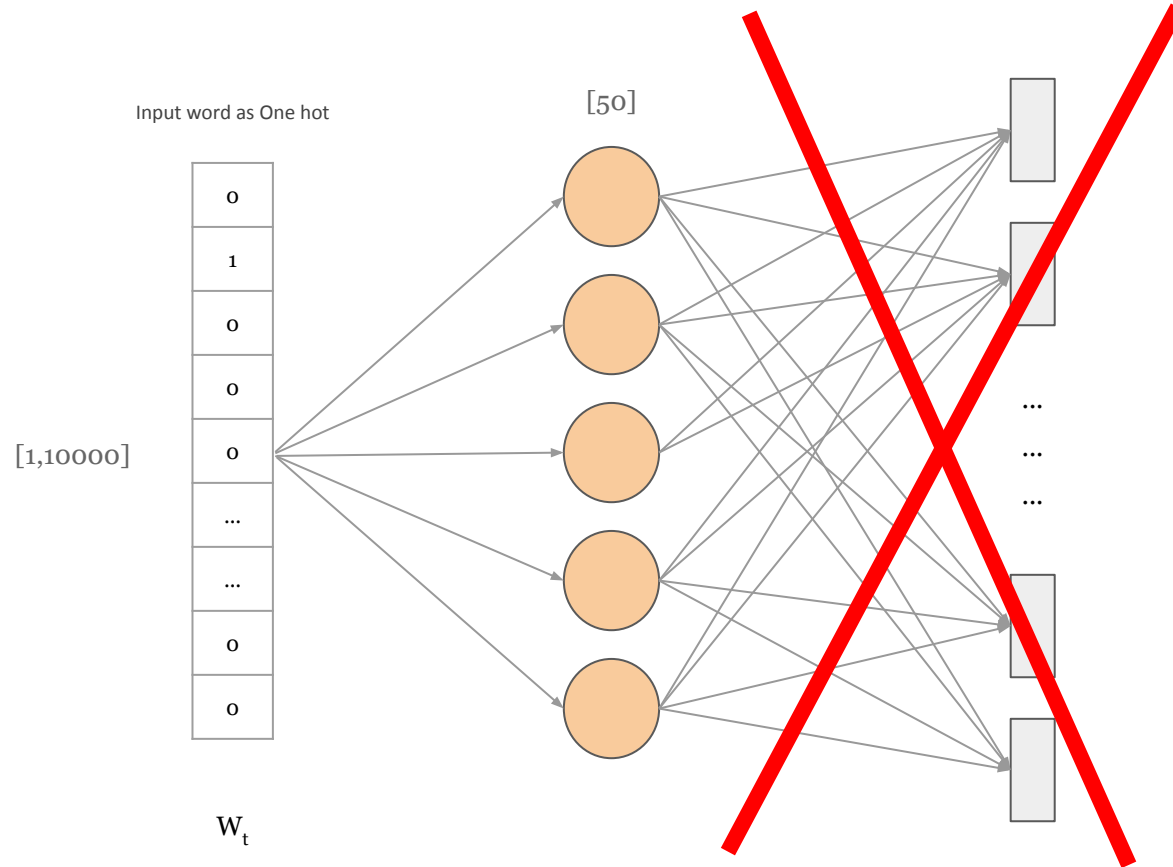


10,000
Predictions are a lot...how
do we handle it better?

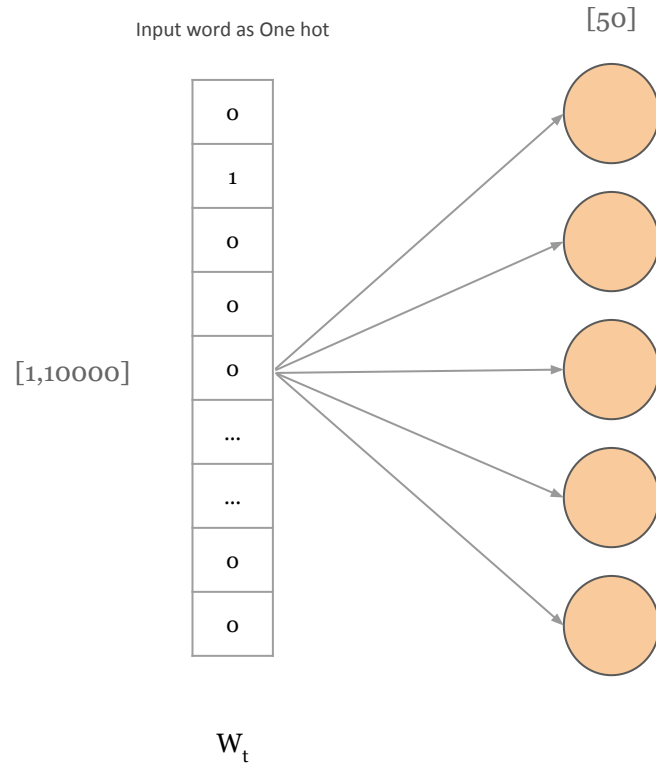




How do we get the word embeddings?

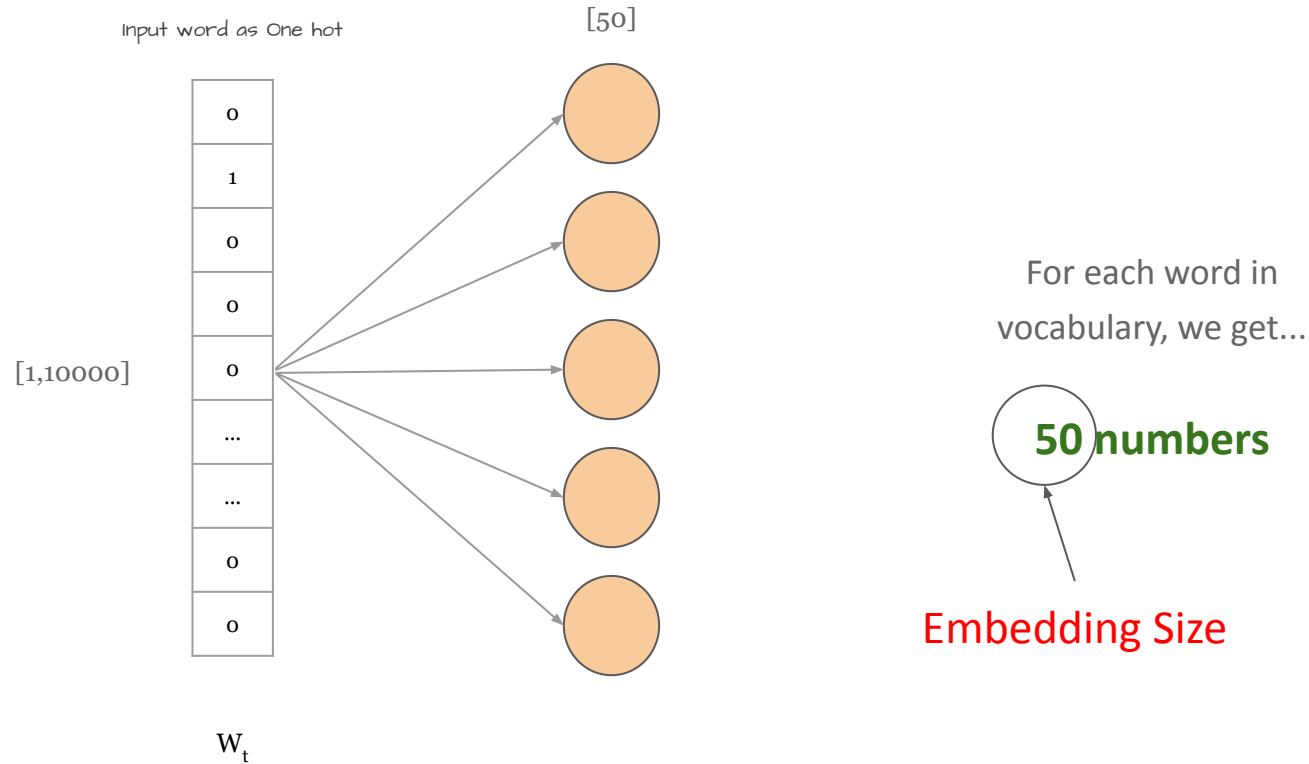


After training
output layer is
discarded



Output of hidden layer is the
word embeddings

For each word in
vocabulary, we get... **50**
numbers





Building Word2Vec Model

Using gensim

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Variations of Word2Vec model

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Global Vectors (GloVe)

1. Similar to Word2Vec in looking at neighbours of a word
2. But also takes into account how many times two words **were neighbours**
3. Approach provided by Stanford University (2014)
4. Lot's of pre-trained models (with different embedding size) available



FastText (by facebook)

1. Very similar to Word2Vec in looking at neighbours of a word
2. Word embedding not only at word level but at **subword** level
3. Subwords are created as character n-grams
4. Can handle **unseen or rare** words

Possible dictionary word/subwords for 'awesome'
awesome → awe, wes, eso, som, ome

Pre-trained Word Embedding models

<https://github.com/RaRe-Technologies/gensim-data>

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Using Pre-Trained Word2Vec model

An Example

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Document #1

He is a good boy. She is also good.

Document #2

Radhika is a good person.

Vocabulary

a, also, boy, good, He, is, person, She, Radhika

	a	also	boy	good	He	is	person	She	Radhika
Index	1	2	3	4	5	6	7	8	9

Build vocabulary and assign index to each unique word

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Document #1

He is a good boy. She is also good.



5, 6, 1, 4, 3, 8, 6, 2, 4

Document #2

Radhika is a good person.



9, 6, 1, 4, 7

	a	also	boy	good	He	is	person	She	Radhika
Index	1	2	3	4	5	6	7	8	9

Create Document Vector - Replace each word by it's index

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Document #1

He is a good boy. She is also good.

5, 6, 1, 4, 3, 8, 6, 2, 4

Vector length = 9

Document #2

Radhika is a good person.

9, 6, 1, 4, 7

Vector length = 5

Make document vectors to be same length

1. Document need to be same size before they can be used in ML model
2. We have to first decide what size each document should be e.g 8 words long
3. If a document has more words than chosen size, we will truncate additional words
4. If document has less words than chosen size, we will pad the document with dummy words

Document #1

He is a good boy. She is also good.

5, 6, 1, 4, 3, 8, 6, 2, 4

5, 6, 1, 4, 3, 8, 6, 2

Document after truncation

Document #2

Radhika is a good person.

9, 6, 1, 4, 7

9, 6, 1, 4, 7, 0, 0, 0

Document after padding

Let's make each document 8 words long

Make document vectors to be same length

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Building Document vector with Padding and Truncation

Using TensorFlow Keras

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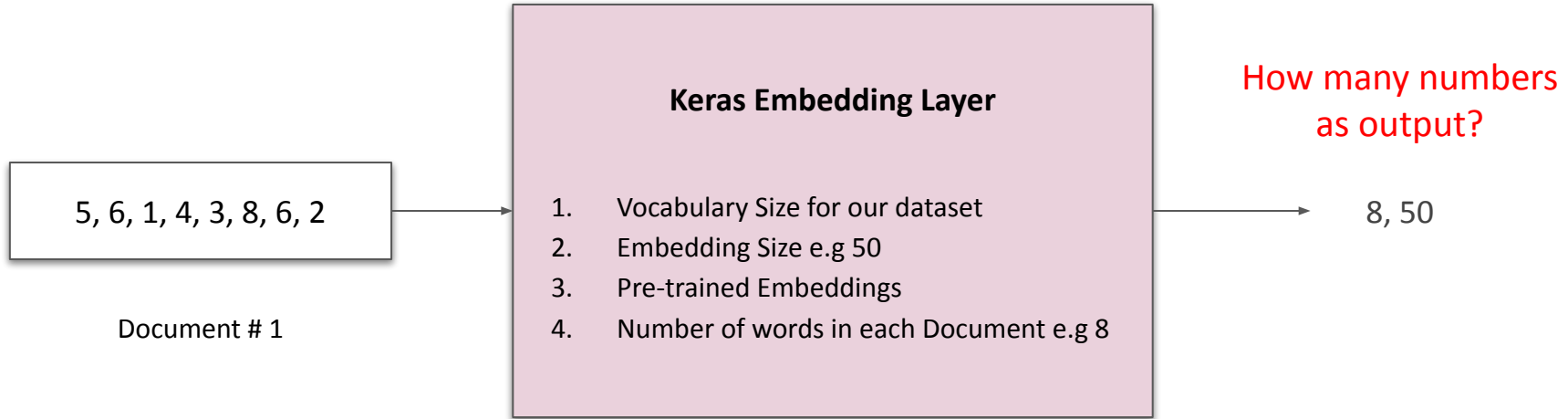
How do we get the word embeddings for our vocabulary words?

Word embeddings from Pre-trained model

1. Download a pre-trained word embedding model
 - a. e.g Google Word2Vec or Glove or fasttext
 - b. Check out <https://github.com/RaRe-Technologies/gensim-data>
 - c. 'Gensim-data' provides an easy way to work with pre-trained word embedding models
2. Extract word embedding for your dataset vocabulary words
 - a. We do not need embedding of all the words in the pre-trained model

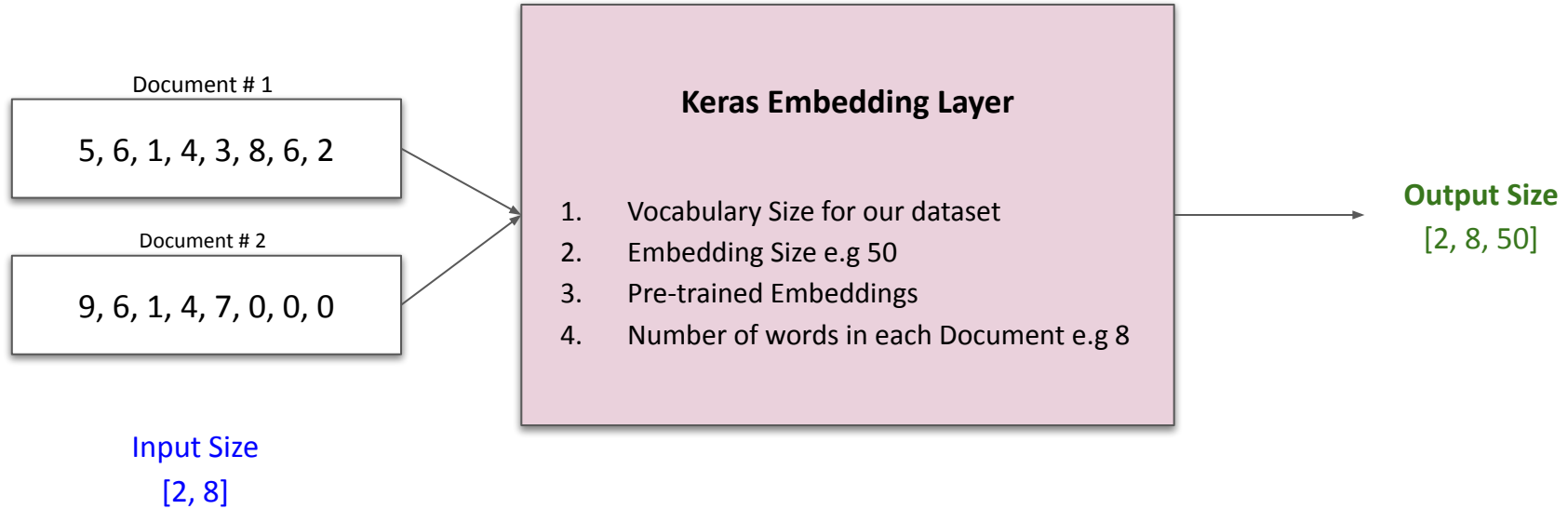


How do we use word embeddings to train a model?



Using Keras Embedding layer

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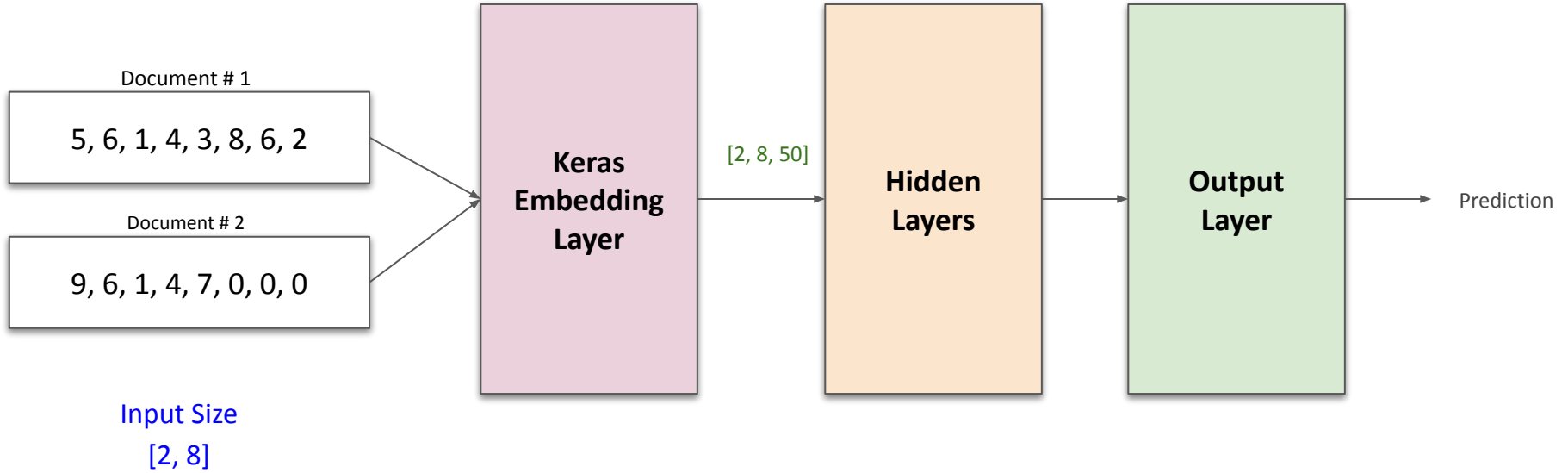
We can feed multiple documents at once *i.e* a batch of documents

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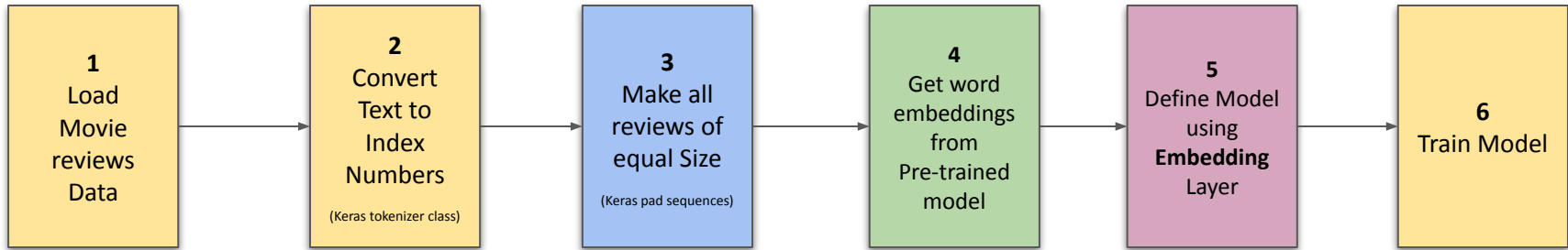


What to do with Embedding layer output?



Feed it to next layer in the model

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Building a model using pre-trained Word Embedding model

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Text Similarity

How close two chunks of texts are by meaning

Web Search

Comparing search query
with document text

Chatbots

Getting answers to
frequently asked
questions

Recommendation

Finding items with
similar descriptions

Text Similarity

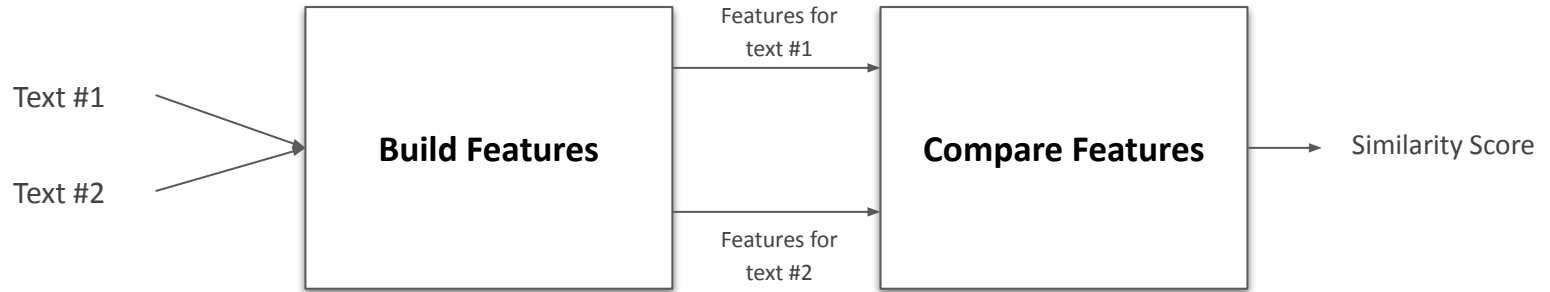
Where it can be useful?

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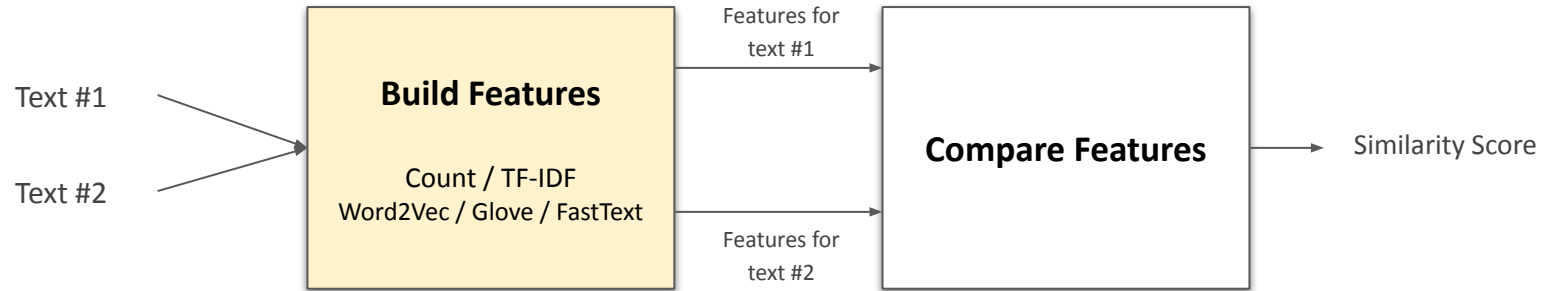
How to check Text Similarity?



Text Similarity

How to do?

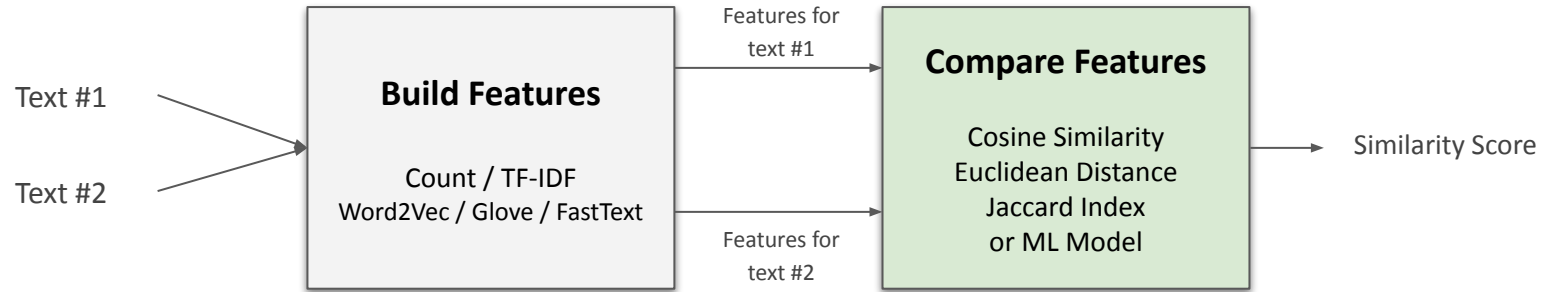
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Text Similarity

How to do?

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Text Similarity

How to do?

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what is topic modeling?

Topic modeling is one of the most common applications of NLP.

we're given a large collection of documents, asked to “make sense” out of it

Popular topic modeling algorithms

1. **Latent Dirichlet allocation (LDA),**
2. Latent semantic analysis (LSA),
3. Probabilistic latent semantic analysis (PLSA).

What does LDA do?

Input:

- D1: I like to eat broccoli and bananas.
- D2: I ate a banana and salad for breakfast.
- D3: Puppies and kittens are cute.
- D4: My sister adopted a kitten yesterday.
- D5: Look at this cute hamster munching on a piece of broccoli.

What does LDA do?

Output:

- Topic A: 30% broccoli, 15% bananas, 10% breakfast, 10% munching
- Topic B: 20% puppies, 20% kittens, 20% cute, 15% hamster

- Document 1 and 2: 100% Topic A
- Document 3 and 4: 100% Topic B
- Document 5: 60% Topic A, 40% Topic B